KKT conditions
Background material

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Basic notions

Consider the problem

$$\begin{aligned} &\min_{x \in \mathcal{X}} f(x) \\ \text{subject to } & g_i(x) \leq 0,\ i = 1, \ldots, m, \\ & h_j(x) = 0,\ j = 1, \ldots, r, \end{aligned}$$

where $\mathcal{X} \subseteq \mathbb{R}^n$ is the ground set on which f, g and h are defined (often $\mathcal{X} = \mathbb{R}^n$; it may also hold the constraints we prefer not to dualize).
The feasible set is

$$\mathcal{F} = \{x \in \mathcal{X} \,|\, g_i(x) \leq 0,\ i = 1, \ldots, m,\ h_j(x) = 0,\ j = 1, \ldots, r\}.$$

x^* is a global minimizer if $\forall x \in \mathcal{F}$, $f(x^*) \leq f(x)$.
x^* is a local minimizer if $\exists \epsilon > 0$ such that $\forall x \in \mathcal{B}(x^*, \epsilon) \cap \mathcal{F}$, $f(x^*) \leq f(x)$.

Lagrangian and Lagrangian dual function

We define the Lagrangian as

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{r} \mu_j h_j(x),$$

and the dual Lagrangian function

$$\mathcal{L}(\lambda, \mu) = \inf_{x \in \mathcal{X}} L(x, \lambda, \mu),$$

the infimum being taken over the ground set, and possibly equal to $-\infty$.

Lagrange multipliers: equality constraints

Consider the mathematical program

$$\min_{x \in \mathcal{X}} f(x)$$
$$\text{subject to } h_j(x) = 0,\ j = 1, \ldots, r,$$

where $\mathcal{X} \subset \mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}$, $h_j : \mathbb{R}^n \to \mathbb{R}$, $j = 1, \ldots, r$.

The Lagrangian of this problem is obtained by associating a Lagrange multiplier μ_j to each constraint function h_j:

$$L(x, \mu) = f(x) + \sum_{j=1}^{r} \mu_j h_j(x).$$

We can obtain very general conditions under which x^* is an optimal solution to the optimization problem, while only basic assumptions are made over $\mathcal{X}$ and the functions f and h_j, $j = 1, \ldots, r$.

Optimality

Theorem

Assume that the Lagrangian associated to the problem

$$\begin{aligned} &\min_{x \in \mathcal{X}} f(x) \\ &s.t.\ \ h_j(x) = 0,\ j = 1, \ldots, r, \end{aligned}$$

has a local minimizer $x^* \in \mathcal{X}$ when the multiplier vector μ is equal to μ^*. If $h_j(x^*) = 0$, $j = 1, \ldots, r$, then x^* is a local minimizer of the constrained problem.

Optimality

Proof.

Assume by contradiction that x^* is not a local minimizer of the constrained problem. Then $\forall \epsilon > 0$, $\exists \overline{x} \in \mathcal{X} \cap \mathcal{B}(x^*, \epsilon)$ such that $h_j(\overline{x}) = 0$, $j = 1, \ldots, r$, and $f(\overline{x}) < f(x^*)$. Thus, $\forall \mu$,

$$\sum_{j=1}^{r} \mu_j h_j(x^*) = \sum_{j=1}^{r} \mu_j h_j(\overline{x}) = 0,$$

and

$$f(\overline{x}) + \sum_{j=1}^{r} \mu_j h_j(\overline{x}) < f(x^*) + \sum_{j=1}^{r} \mu_j h_j(x^*).$$

Taking $\mu = \mu^*$, the previous inequality contradicts that x^* is a local minimizer of the Lagrangian when $\mu = \mu^*$. □

Lagrange multipliers: inequality constraints

Same setting, with inequality constraints: $\mathcal{X} \subset \mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}$, $g_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, m$.

Theorem

Assume that the Lagrangian associated to the problem

$$\begin{aligned} &\min_{x \in \mathcal{X}} \ f(x) \\ &s.t. \ \ g_i(x) \leq 0, \ \ i = 1, \ldots, m, \end{aligned}$$

has a local minimizer $x^* \in \mathcal{X}$ when the multiplier vector λ is equal to λ^*. If $g_i(x^*) \leq 0$, $\lambda_i^* \geq 0$, and $\lambda_i^* g_i(x^*) = 0$, $i = 1, \ldots, m$, then x^* is a local minimizer of the constrained problem.

Lagrange multipliers: inequality constraints

Proof.

As previously, assume by contradiction that x^* is not a local minimizer of the constrained problem. Then $\forall \epsilon > 0$, $\exists\, \overline{x} \in \mathcal{X} \cap \mathcal{B}(x^*, \epsilon)$ such that $g_i(\overline{x}) \leq 0$, $i = 1, \ldots, m$, and $f(\overline{x}) < f(x^*)$. Therefore, for $\lambda = \lambda^* \geq 0$,

$$\sum_{i=1}^{m} \lambda_i g_i(\overline{x}) \leq 0 \text{ and } \sum_{i=1}^{m} \lambda_i g_i(x^*) = 0.$$

Consequently,

$$f(\overline{x}) + \sum_{i=1}^{m} \lambda_i g_i(\overline{x}) < f(x^*) + \sum_{i=1}^{m} \lambda_i g_i(x^*),$$

contradicting that x^* is a local minimizer of the Lagrangian when $\lambda = \lambda^*$. □

Dual problem

The dual problem is

$$\max_{\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}^r} \mathcal{L}(\lambda, \mu)$$
$$\text{such that } \lambda \geq 0.$$

Important properties:

- The dual problem is always convex, i.e. $\mathcal{L}$ is always concave (even if the primal problem is not convex).
- The primal and dual (global) optimal values, f^* and $\mathcal{L}^*$, always satisfy the weak duality: $f^* \geq \mathcal{L}^*$.
- Strong duality: under some conditions (constraint qualifications), $f^* = \mathcal{L}^*$.

Saddle point

Since

$$\max_{\lambda \geq 0,\ \mu} L(x, \lambda, \mu) = \begin{cases} f(x) & \text{if } x \in \mathcal{F}, \\ +\infty & \text{otherwise,} \end{cases}$$

the primal problem reads $\min_{x \in \mathcal{X}} \max_{\lambda \geq 0,\ \mu} L(x, \lambda, \mu)$, and the dual problem $\max_{\lambda \geq 0,\ \mu} \min_{x \in \mathcal{X}} L(x, \lambda, \mu)$: weak duality is nothing but $\max \min \leq \min \max$.

Definition (Saddle point)

(x^*, λ^*, μ^*), *with* $\lambda^* \geq 0$, *is a saddle point of* L *if*

$$L(x^*, \lambda, \mu) \leq L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \quad \forall\, x \in \mathcal{X},\ \forall\, \lambda \geq 0,\ \forall\, \mu.$$

(x^*, λ^*, μ^*) is a saddle point if and only if x^* is primal optimal, (λ^*, μ^*) is dual optimal, and the duality gap is null.

Working with the dual function

$\mathcal{L}$ is concave, being an infimum of functions that are affine in (λ, μ), but it is usually not differentiable. A supergradient is however immediate.

Theorem

Let $\overline{x} \in \arg\min_{x \in \mathcal{X}} L(x, \lambda, \mu)$. *Then, for all* (λ', μ'),

$$\mathcal{L}(\lambda', \mu') \leq \mathcal{L}(\lambda, \mu) + g(\overline{x})^T(\lambda' - \lambda) + h(\overline{x})^T(\mu' - \mu).$$

Proof.

$\mathcal{L}(\lambda', \mu') \leq L(\overline{x}, \lambda', \mu') = L(\overline{x}, \lambda, \mu) + g(\overline{x})^T(\lambda' - \lambda) + h(\overline{x})^T(\mu' - \mu)$, and $L(\overline{x}, \lambda, \mu) = \mathcal{L}(\lambda, \mu)$ by definition of $\overline{x}$. □

The constraint values at the Lagrangian minimizer thus give a supergradient, the search direction of the dual ascent methods: $\lambda^{\nu+1} = \max\{\lambda^\nu + \alpha^\nu g(\overline{x}^\nu), 0\}$, the maximum projecting on $\lambda \geq 0$. As $\mathcal{L}$ is non-smooth, such a direction is not necessarily an ascent one: the step α^ν must be chosen accordingly.

Duality gap

Given a primal feasible solution x and a dual feasible solution (λ, μ), the quantity $f(x) - \mathcal{L}(\lambda, \mu)$ is called the duality gap between x and (λ, μ). Note that

$$f(x) - f^* \leq f(x) - \mathcal{L}(\lambda, \mu).$$

Therefore, if the duality gap is equal to 0, then x is primal-optimal (and similarly, λ and μ are dual-optimal).

From an algorithmic point of view, if strong duality holds, this provides a stopping criterion: if $f(x) - \mathcal{L}(\lambda, \mu) \leq \epsilon$, then $f(x) - f^* \leq \epsilon$.

Duality gap: local case

We can also define the dual Lagrangian function restricted to the ball $\mathcal{B}(x^*, \epsilon)$:

$$\mathcal{L}_{\mathcal{B}(x^*,\epsilon)}(\lambda, \mu) = \min_{x \in \mathcal{B}(x^*,\epsilon)} L(x, \lambda, \mu).$$

The weak duality still holds locally:

$$\mathcal{L}^*_{\mathcal{B}(x^*,\epsilon)} \leq f(x^*).$$

Under some conditions, the strong duality also holds:

$$\mathcal{L}^*_{\mathcal{B}(x^*,\epsilon)} = f(x^*).$$

Duality gap: local case

Note however that

$$\min_{x \in \mathbb{R}^n} L(x, \lambda, \mu) \leq \min_{x \in \mathcal{B}(x^*, \epsilon)} L(x, \lambda, \mu)$$

so

$$\mathcal{L}^* \leq \mathcal{L}^*_{\mathcal{B}(x^*, \epsilon)}.$$

Therefore, if x^* is a local minimizer and the strong duality locally holds,

$$\mathcal{L}^* \leq f(x^*).$$

The inequality can be strict.

Karush-Kuhn-Tucker (KKT) conditions

Consider $f, g_i, h_j \in C^1$, $i = 1, \ldots, m$, $j = 1, \ldots, r$, and the problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \ & f(x) \\ \text{s.t.} \ & g_i(x) \leq 0, \ i = 1, \ldots, m, \\ & h_j(x) = 0, \ j = 1, \ldots, r. \end{aligned}$$

Karush-Kuhn-Tucker (KKT) conditions:

$$\begin{aligned} \nabla_x L(x, \lambda, \mu) &= 0 && \text{(stationarity)} \\ \lambda_i g_i(x) &= 0 && \text{(complementarity)} \\ g_i(x) \leq 0, \ h_j(x) &= 0 \ \forall i, j && \text{(primal feasibility)} \\ \lambda_i &\geq 0 \ \forall i && \text{(dual feasibility)} \end{aligned}$$

Necessary conditions

Let x^* be a minimizer of f over $\mathcal{F} \cap \mathcal{B}(x^*, \epsilon)$, $\epsilon > 0$, and (λ^*, μ^*) be a dual solution if x is restricted to $\mathcal{B}(x^*, \epsilon)$, with a zero duality gap (the strong duality holds). Then

$$\begin{aligned} f(x^*) &= \mathcal{L}_{\mathcal{B}(x^*,\epsilon)}(\lambda^*, \mu^*) \\ &= \min_{x \in \mathcal{B}(x^*,\epsilon)} \left(f(x) + \sum_{i=1}^{m} \lambda_i^* g_i(x) + \sum_{j=1}^{r} \mu_j^* h_j(x) \right) \\ &\leq f(x^*) + \sum_{i=1}^{m} \lambda_i^* g_i(x^*) + \sum_{j=1}^{r} \mu_j^* h_j(x^*) \\ &\leq f(x^*) \end{aligned}$$

Thus, x^* is a minimizer of $L(x, \lambda^*, \mu^*)$ in $\mathcal{B}(x^*, \epsilon)$, and $\nabla_x L(x^*, \lambda^*, \mu^*) = 0$.

We have obtained the stationarity conditions.

Necessary conditions

The previous inequalities also imply $\sum_{i=1}^{m} \lambda_i^* g_i(x^*) = 0$ as $\sum_{i=1}^{m} \lambda_i^* g_i(x^*) \leq 0$, and consequently, $\lambda_i^* g_i(x^*) = 0, \ \forall i$.

This establishes the complementarity conditions.

If x^* is a global minimizer, we can replace $\mathcal{B}(x^*, \epsilon)$ by $\mathbb{R}^n$.

We can summarize our findings in the theorem below.

Theorem (KKT necessary conditions)

If x^*, (λ^*, μ^*) are primal and dual solutions with a null duality gap, then x^*, (λ^*, μ^*) satisfy the KKT conditions.

Strong duality

The strong duality assumption often plays a key role. How to ensure that it holds?

- Linear programming: it always holds, as soon as one of the two programs has a finite optimal value.
- Convex programming: the program is convex when $\mathcal{X}$ is convex, f and the g_i are convex, and the h_j are affine. The Slater condition is then sufficient: there exists a feasible x in the relative interior of $\mathcal{X}$ with $g_i(x) < 0$ for every non-affine g_i.
- Nonconvex programming: a constraint qualification is required. The most common, and the most demanding, is the linear independence constraint qualification (LICQ).

Nonconvex programming

Theorem (Necessary conditions)

If x^* is a local solution of

$$\begin{aligned} \min_{x \in \mathcal{X}} \ & f(x) \\ s.t. \ & g_i(x) \leq 0, \ i = 1, \ldots, m \\ & h_j(x) = 0, \ j = 1, \ldots, r, \end{aligned}$$

where f, the g_i, $i = 1, \ldots, m$, and the h_j, $j = 1, \ldots, r$, are C^1, and a constraint qualification condition holds at x^*, then $\exists (\lambda^*, \mu^*)$ such that the KKT conditions hold at (x^*, λ^*, μ^*).

Proof.

See Nocedal & Wright, "Numerical Optimization", Section 12.4. □

Sufficiency of KKT conditions

Let (x^*, λ^*, μ^*) satisfy the KKT conditions. Primal feasibility and complementarity give

$$L(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_{i=1}^{m} \lambda_i^* g_i(x^*) + \sum_{j=1}^{r} \mu_j^* h_j(x^*) = f(x^*).$$

In the convex case, $L(\cdot, \lambda^*, \mu^*)$ is convex, so stationarity makes x^* a *global* minimizer of the Lagrangian over $\mathcal{X}$:

$$\mathcal{L}(\lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*) = f(x^*).$$

The duality gap is null (strong duality) and x^*, (λ^*, μ^*) are globally optimal for the primal and the dual, respectively.

In the nonconvex case, stationarity only makes x^* a critical point of $L(\cdot, \lambda^*, \mu^*)$: it may be a local minimizer, a saddle point, or even a maximizer, and $\mathcal{L}(\lambda^*, \mu^*) < f(x^*)$ remains possible. The KKT conditions are then necessary, under a constraint qualification, but not sufficient — not even locally, as the next slide shows.

KKT and the local duality gap

The necessary conditions were obtained from the local strong duality. Does the converse hold: do the KKT conditions restore a null gap on a small enough ball?

As $L(x^*, \lambda^*, \mu^*) = f(x^*)$, the local gap vanishes on $\mathcal{B}(x^*, \epsilon)$ if and only if x^* *minimizes* $L(\cdot, \lambda^*, \mu^*)$ on that ball. The KKT conditions only provide stationarity, which is strictly weaker. A constraint qualification cannot help: it is what makes the conditions *necessary* at a local solution, and says nothing about what they imply.

Counterexample

$\min_{x \in \mathbb{R}^2} x_1^2 - x_2^2$ subject to $h(x) = x_2 = 0$. The feasible set is the axis $x_2 = 0$, on which $f = x_1^2$: $x^* = (0, 0)$ is the unique global solution, and $L(x, \mu) = x_1^2 - x_2^2 + \mu x_2$ is stationary at x^* for $\mu^* = 0$. Regularity is not at fault: $\nabla h = (0, 1) \neq 0$, so LICQ holds, and on the critical cone $\{d \mid d_2 = 0\}$, which is not trivial, $\nabla^2_{xx} L = \mathrm{diag}(2, -2)$ is positive definite — the second-order sufficient conditions hold too.

KKT and the local duality gap (cont'd)

Yet $L(\cdot, \mu^*) = x_1^2 - x_2^2$ has a saddle point at x^*, so for *every* $\epsilon > 0$

$$\mathcal{L}_{\mathcal{B}(x^*,\epsilon)}(\mu^*) = \min_{\|x\| \leq \epsilon} \left(x_1^2 - x_2^2\right) = -\epsilon^2 < 0 = f(x^*).$$

The gap does tend to 0 with ϵ, which makes the mistake tempting, but it vanishes on no neighbourhood. And the failure has nothing to do with the optimality of x^*, which is global: it comes from the behaviour of the Lagrangian itself.

A sufficient condition is $\nabla^2_{xx} L \succ 0$ on the whole space, and not only on the critical cone: above, $\mathrm{diag}(2, -2)$ is positive definite on the cone but indefinite on $\mathbb{R}^2$.

Restoring a local zero gap

The augmented Lagrangian is precisely the standard remedy:

$$L_\rho(x, \lambda, \mu) = L(x, \lambda, \mu) + \frac{\rho}{2}\|h(x)\|^2,$$

whose Hessian at x^* is $\nabla^2_{xx} L + \rho \nabla h(x^*) \nabla h(x^*)^T$. Under LICQ and the second-order conditions, it becomes positive definite for ρ large enough, so x^* locally minimizes L_ρ and the local gap vanishes.

On the previous counterexample, $L_\rho(x, 0) = x_1^2 + (\rho/2 - 1)x_2^2$, of Hessian $\mathrm{diag}(2, \rho - 2)$: the gap is positive for $\rho \leq 2$ and null as soon as $\rho > 2$.

This is the motivation for the augmented Lagrangian methods, and in stochastic programming for progressive hedging, where the penalty is placed on the nonanticipativity constraints.

Active set

Definition (Active set)

The active set $\mathcal{A}(x)$ of the optimization problem

$$\begin{aligned} \min_{x \in \mathcal{X}} \ & f(x) \\ s.t. \ & g_i(x) \leq 0, \ i \in \mathcal{I} \\ & h_i(x) = 0, \ i \in \mathcal{E}, \end{aligned}$$

in a feasible point x is the index set of the equality constraints and the active inequality constraints at that point:

$$\mathcal{A}(x) = \mathcal{E} \cup \{ i \in \mathcal{I} \,|\, g_i(x) = 0 \}$$

LICQ

The most popular constraint qualification is the LICQ.

Definition (LICQ)

Given a feasible point x and its active set $\mathcal{A}(x)$, the linear independence constraint qualification (LICQ) holds at x if the gradients of the active constraints, $\{\nabla_x h_i(x),\ i \in \mathcal{E}\} \cup \{\nabla_x g_i(x),\ i \in \mathcal{A}(x) \cap \mathcal{I}\}$, are linearly independent.